4

How Do We Know It's Working?

Evidence in Support of Inclusive Education

There is an increased focus on educating students with disabilities in inclusive general education settings due to converging policy guidelines, ethics, and research-based evidence over the past several decades. As discussed in Chapter 1, the 2004 Individuals With Disabilities Education Act (IDEA) and the 2015 Every Student Succeeds Act focus on students learning the general education curriculum, in the general education setting. Moreover, IDEA requires schools to provide students with disabilities "access to the general education curriculum . . . to learn grade-level content based on grade-level standards" (§300.26(b)(3)(ii)), whereby the state standards determine the core curriculum.

In addition to policy reasons, inclusive education is supported by ethical considerations. One key ethical rationale is that disability is simply part of human diversity and the human experience. This view of disability encompasses acceptance and respect. It also highlights that, for a community to embrace and celebrate the diversity within each person, we must learn together and from one another to know and appreciate one another and move beyond simply tolerance of one another. Inclusive education, as such, is fundamental to achieving and celebrating diversity, as it advocates that students learn together in schools to prepare them to live and work together throughout life.

A growing and substantial body of research supports inclusive education; in fact, the past 40 years of research shows that, when students with and without disabilities learn together, they experience better outcomes. A key benefit of inclusive settings is the presence of the general education teacher, who has expertise in the core academic content, along with materials and tools for teaching that content (Kleinert et al., 2015). Inclusive education also has been shown to improve the ability of teachers to provide quality instruction to all students, including students with disabilities (Finke et al., 2009). Other benefits of inclusive education for students with disabilities include improvements to quality of life and enhanced educational experiences (Hunt et al., 2012). Research further suggests inclusive education contributes to improved postschool outcomes for students with disabilities (Mazzotti et al., 2013; Test et al., 2009).

This chapter focuses on describing the research evidence supporting inclusive education across five domains: academic, social-emotional learning, communication, friendships and relationships, and the impact on nondisabled students. We then consider how these data compel us to make changes within schools to provide opportunities for people with disabilities to experience positive outcomes across the life span. We consider two groups of students with disabilities: (a) Students with "high-incidence" disabilities comprise the majority of students with disabilities, which include students with learning disabilities, attention deficit disorders, and speech and language impairments, among others; (b) Students with "low-incidence disabilities" comprise about 1% of students who have significant cognitive disability and are eligible to take their state's alternate assessment.

Academic Progress

Educational research indicates that increased academic competency for *all* students in the areas of literacy, math, and science is associated with improved future outcomes. For example, literacy skills lead to substantially improved opportunities for educational, economic, and overall life outcomes. In the area of mathematics, even the most basic knowledge of math skills increases employability prospects later in life. Competency in higher-level math skills leads to additional postsecondary educational opportunities, a variety of potential career options, and increased economic success (Wei et al., 2012).

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Clearly, academic achievement matters and should be considered when developing individualized education programs (IEPs) for students with disabilities, and when determining where those IEP services are provided. A large and growing body of research demonstrates that students with disabilities learn academic skills equally, if not better, in inclusive settings compared to self-contained special education classrooms. We first consider students with high-incidence disabilities. Several studies have directly compared outcomes for students with high-incidence disabilities taught in inclusive settings versus students taught in separate special education or resource classrooms. For example, Tremblay (2013) compared the reading and writing scores of students with learning disabilities taught in inclusive versus self-contained classrooms and found students in inclusive classrooms obtained higher scores. Similarly, Cole et al. (2020) compared the state assessment reading and math scores of nearly 1,700 students in fourth through eighth grade with high-incidence disabilities. Students with disabilities who spent more time in general education settings did significantly better in both math and reading assessments than did students who spent more time in separate special education classrooms.

A similar pattern of academic benefit is present for students with low-incidence disabilities. For example, Dessemontet et al. (2012) investigated the literacy and math outcomes of 68 children with intellectual disability: 34 were taught in an inclusive school, and 34 were taught in separate special education classrooms. The students in the inclusive classroom had better literacy skills, and there was no difference in math outcomes. Kurth and Mastergeorge (2010) reviewed the educational progress of 15 students with autism and intellectual disability, whose educational records were reviewed from kindergarten through ninth grade. The students had been continuously enrolled in either an inclusive school or separate special education classroom for 10 years (K-9). The researchers found those students who had been taught in inclusive classrooms obtained significantly higher reading, math, and writing outcomes than their matched peers in self-contained programs.

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Social-Emotional Learning

Often, students with disabilities are removed from general education classrooms because they engage in problem behaviors. Sometimes students struggle to learn because of unmet social-emotional needs, thus putting them at risk for removal to remedial special education programs or classrooms dedicated to teaching students with emotional-behavioral disorders. Other times, teachers have difficulty connecting with students who have problem behaviors or who are withdrawn, resulting in students falling behind and being referred for special education services. Clearly, social, emotional, and behavioral outcomes are important to helping students learn and thrive at school. Research has demonstrated students have more positive social-emotional trajectories into adulthood, engaging in fewer antisocial behaviors, when they are taught in inclusive settings (Woodman et al., 2016). This research suggests that removal of students due to problem behavior is counterproductive, and students should be taught emotional regulation skills in context. In other words, students should be taught coping strategies and/or replacement behaviors in the natural setting, when and where those skills are needed.

One strategy for teaching social-emotional skills in the inclusive context is using peer-assisted learning; this has been found to be highly effective in promoting positive social-emotional outcomes for students with and at risk for experiencing disability (Moeyaert et al., 2019). Another consideration related to social-emotional learning is related to student self-sufficiency and independence, which includes such skills as being able to do tasks independently. A longitudinal study found that students with disabilities taught in inclusive settings demonstrated more independence and self-sufficiency than did students who were taught in self-contained special education classes (Newman & Davies-Mercier, 2005). Students taught in inclusive settings demonstrate better self-determination skills overall (Hughes et al., 2013). Finally, researchers have found that adolescents with disabilities taught in inclusive settings are more engaged in curricular activities than their peers who are taught in self-contained settings (Kurth & Mastergeorge, 2012). Together, these research studies demonstrate that inclusive education is strongly associated with a variety of positive social-emotional outcomes for students with disabilities.

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Communication

Communication is at the core of the school experience, as it is necessary to engage in academic learning with teachers and to socialize with peers. In fact, communication is used throughout the day for nearly every activity, both as children and later as adults. For those reasons, communication is frequently mentioned as a critical personalized education skill. But for students with disabilities who have complex communication needs—students who need supports to communicate effectively—this experience is compromised when communication needs are not addressed. Students with complex communication needs may use such supports as pictures or text to supplement their voice, whereas some students communicate with augmentative and alternative communication, such as speech-generating devices. Students with disabilities who have complex communication needs are at risk of not gaining access to the general education curriculum, obtaining poor academic outcomes, not being placed in the general education classroom, and experiencing low expectations from school personnel (Kurth et al., 2014; Kleinert et al., 2015; Ruppar et al., 2011).

Just as students with disabilities who demonstrate problem behaviors are more likely to be taught in self-contained settings, so too are students with disabilities who have communication support needs. Although far too many students with complex communication needs leave school without an effective means of communicating, inclusive education has been shown to improve communication outcomes (Foreman et al., 2016). One reason for this is that inclusive settings are far more likely to contain communication partners. Within self-contained classrooms, many students have complex communication needs, leaving fewer competent communication partners with whom to practice and develop social communication skills. Inclusive settings also contain more students, meaning there are better chances of finding peers with similar interests with whom to communicate. Finally, the discourse of inclusive classrooms (the language students are exposed to) has been found to be much richer, giving students more opportunities to hear varied language structures and develop communication skills.

Friendships and Relationships

Ask anyone what they most remember about their time in school, and invariably they think about the social activities they enjoyed with friends, either at recess in elementary school or in sports, clubs, or other after school events in secondary school. Forming friendships and relationships is crucial to development and, frankly, for many results in some of the most enjoyable aspects of school and work. Given the importance of social relationships, friendship and relationship outcomes of inclusive education have been extensively researched. This research has consistently found that students with disabilities taught in inclusive settings have better social skills and more robust social networks than do students with disabilities taught in separate classrooms (Fisher & Meyer, 2002).

Researchers have suggested that students with disabilities have greater growth in social skills when they are taught in inclusive settings, largely as a result of access to social networks and peer models (McDonnell et al., 2002). In fact, students with disabilities taught in inclusive settings have larger social networks and more opportunities to develop social skills (Kennedy & Itkonen, 2016). Students with disabilities taught in inclusive settings have higher levels of social engagement (Lyons et al., 2011).

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Impact on Students Without Disabilities

So far in this chapter we have focused on students with disabilities; this research has highlighted the many benefits of inclusive education for students with disabilities. However, many teachers, parents, and administrators are concerned about the impact of inclusive education on students without disabilities. What do we know about the outcomes of inclusive education for students who do not have a disability? Luckily, a large body of research dedicated to this question has demonstrated that educating students with disabilities within the general education classroom does not harm

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students without disabilities and may benefit them. In 2007 researchers systematically reviewed 26 different studies focusing on the impact of inclusive education on the academic development of students without disabilities, finding that 81% of study outcomes reported students without disabilities either experienced no effects or experienced positive effects (Kalambouka et al., 2007). A similar review completed in 2009 came to the same conclusion: inclusive education of students with disabilities has either a positive or neutral impact on the academic outcomes of students without disabilities (Ruijs & Peetsma, 2009).

Why might inclusive education have either a positive or neutral impact on learners without disabilities? Several reasons are possible. First, teachers in inclusive classrooms are likely to adapt how they teach to benefit all students, including the use of strategies and techniques that meet the needs of diverse learners (Dessemontet & Bless, 2013). Second, teachers working in inclusive settings may collaborate more, gathering strategies and supports from colleagues in ways that benefit all students (Sharma et al., 2008). Third, coordinated schoolwide approaches to positive behavior support and multitiered systems of supports may also equip teachers with strategies to meet the behavior and learning needs of all students through a collaborative, interdisciplinary problem-solving approach (Giangreco et al., 1993).

Examining Trends in Placement

Research spanning several decades demonstrates students with and without disabilities benefit from an inclusive education, which also supports effective teaching. We also note that, when asked, parents and families tend to support an inclusive education, believing it benefits their children academically and socially (Peck et al., 2004). Considering that IDEA requires education teams to make educational decisions based on research, it is clear the research supports inclusive education, and this should therefore be the de facto placement for students with disabilities who receive special education services in the United States. In short, we are out of excuses *not* to include students with disabilities. There is no compelling reason for segregation and separation for part or all of the school day.

However, review of data reported by states on the implementation of IDEA, which includes where students with disabilities receive special education services, reveals limited progress in teaching all students with disabilities in the inclusive setting. According to the most recent report to Congress on the implementation of IDEA (Office of Special Education and Rehabilitative Services, 2019), only 63.5% of students with disabilities are included in general education settings; 18.1% were taught in resource settings (part of the day in general education and part of the day in separate classrooms), 13.3% were taught mostly in separate self-contained classrooms, and 5% were taught in entirely separate schools. A closer look at these data reveal troubling trends: little progress has been made in educating more students with disabilities in inclusive settings, and students with high-incidence disabilities are much more likely to be included in general education compared to students with low-incidence disabilities.

In considering progress in educating more students with disabilities in inclusive settings, McLeskey et al. (2012) tracked student placement between the 1990–1991 school year and the 2007–2008 school year. They found that approximately 39% of students with disabilities were taught in inclusive settings in the 1990–1991 school year, and 65.5% were taught in this placement by the 2007–2008 school year. However, most of this increasing placement in inclusive settings came from students with high-incidence disabilities, specifically students with learning disabilities and speech language impairments. Students with emotional-behavioral disorders experienced very little change. Morningstar, Kurth, and colleagues (2017) completed a similar review focused on students with low-incidence disabilities, examining placement trends between 2000 and 2014 for students with autism, intellectual disability, multiple disability, and deaf-blindness. They found that this group of students is about four times more likely to be taught in separate self-contained classrooms than in general education inclusive settings, representing almost no change between 2000 and 2014.

Pulling It All Together

As we have established in this chapter, placement matters for student outcomes within school. Emerging research also demonstrates that school placement has lifelong consequences. Unfortunately, students with disabilities are at risk for absenteeism and grade retention, as well as not graduating from high school, which has numerous consequences, including poor economic outcomes and increased risk for experiencing lifelong poor health outcomes (Yoder & Cantrell, 2019). The most recent report to Congress on the implementation of IDEA found that, among students with disabilities between the ages of 14 and 21, approximately

15% dropped out of school, only 45% graduated with a high school diploma, and only 2% received a certificate of completion (Office of Special Education and Rehabilitative Services, 2019). Among 18- to 24-year-olds, the dropout rate for people with disabilities is 15.2%, compared to 6.4% for people without disabilities. In fact, people with disabilities are more likely to drop out of school than any other group (National Center for Education Statistics, 2020).

Employment data offer equally bleak outcomes (Sannicandro et al., 2018), with rates of employment for persons with disabilities remaining static at around 34% (Smith et al., 2017). Eighty-one percent of adults with disabilities received segregated services in day habilitation, clinic, rehabilitation, or other segregated day programs as adults (Braddock et al., 2015). Because the Fair Labor Standards Act (29 CFR §14(c)) allows employers to pay workers with disabilities subminimum wages, earnings of employees in sheltered workshops and other segregated settings are significantly reduced (Kregel & Dean, 2002). Of all participants in total day or work programs, only 19% worked in supported employment (Braddock et al., 2015), which substantially increases earnings of workers (Wehman et al., 2018).

Young adults with low-incidence disabilities experience the poorest postsecondary outcomes of any disability group (Newman et al., 2011). Students with low-incidence disabilities continue to exit school without the skills they need to be successful (Kearns et al., 2010). Many students have inadequate communication supports available by the time they leave school (Erickson & Geist, 2016; Kleinert et al., 2015), which can substantially impact postschool employment, social, and recreational opportunities. Limited social skills and lowered expectations are associated with diminished postschool employment outcomes (Carter et al., 2012). For people with low-incidence disabilities, an equally concerning postschool outcome is related to integrated community living. Although people with low-incidence disabilities report preferring to live in apartments with support (Ioanna, 2018), nearly 71% lived with a family or caregiver, and only 16% lived alone or with a roommate in the community (Braddock et al., 2015).

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Together, these findings demonstrate a systemic failure of the current model of teaching students with disabilities (Morningstar, Zagona, et al., 2017). Reviewing the outcomes of special education (e.g., employment, independent living, postsecondary education) clearly demonstrates that the status quo of teaching remedial skills in special education classrooms has never led to positive outcomes for students with disabilities once they leave school. We are thus compelled to do better; we must make changes in schools to better prepare for students for life after school, and this includes educating students in general education classrooms, with the supports they need, to learn the general education curriculum. Preliminary data supports this: students with disabilities who had received special education services in inclusive general education settings were more likely to obtain postsecondary education, be employed, and live independently as adults (Mazzotti et al., 2016; Test et al., 2009).

In summary, the research we have reviewed in this chapter demonstrates that students with disabilities have better outcomes in school when taught in inclusive classrooms. They are more likely to achieve important academic, social, emotional, behavioral, communication, and relationship outcomes when taught alongside their peers without disabilities. Students taught in inclusive settings during their school years also achieved more positive postschool outcomes as adults, including rates of employment, postsecondary education, and independent living. Finally, students without disabilities experienced no negative consequences from being taught in classrooms that included students with disabilities. Thus, these data offer compelling reasons to teach all students, including those with disabilities, the general education curriculum in inclusive settings.

Reflective Questions for Group Study

1. Summarize the research around inclusive education for students with and without disabilities.
2. Describe the extent to which students with high- and low-incidence disabilities learn in inclusive settings, and any notable trends in their placement over time.
3. How does school placement in inclusive or special classes impact long-term outcomes?
4. Is our current model of educating students effective in promoting positive lifelong outcomes? Why or why not?